

Customer number	Arrival time	Service time
1	3	4
2	8	3
3	9	4
4	17	6
5	18	3
6	19	2
7	20	2
8	25	3
9	27	2

- In our simulation, we keep track of a number of variables
 - MC = Master clock
 - Status of
 - Server: 1 = BUSY, 0 = IDLE
 - Buffer: 1 = OCCUPIED, 0 = VACANT
 - Event time:
 - Next arrival event and service time of this arrival
 - Next departure event and arrival time of this departure
 - The (arrival time, service time) of the customer in buffer
 - In order to compute the response time, we keep track of
 - The cumulative response time (T)
 - Cumulative number of customers rejected (R)

	Meaning
λ	Arrival rate
μ	service rate Mean service time = $1 / \mu$
ρ	Utilisation
m	Number of servers
n	Number of holding slots

P-K formula

$$W = \frac{\lambda E[S^2]}{2(1 - \rho)}$$

where $\rho = \lambda / \mu$

Mean response time =
Mean waiting time + mean service time

(Week 1A)

Alice's waiting time

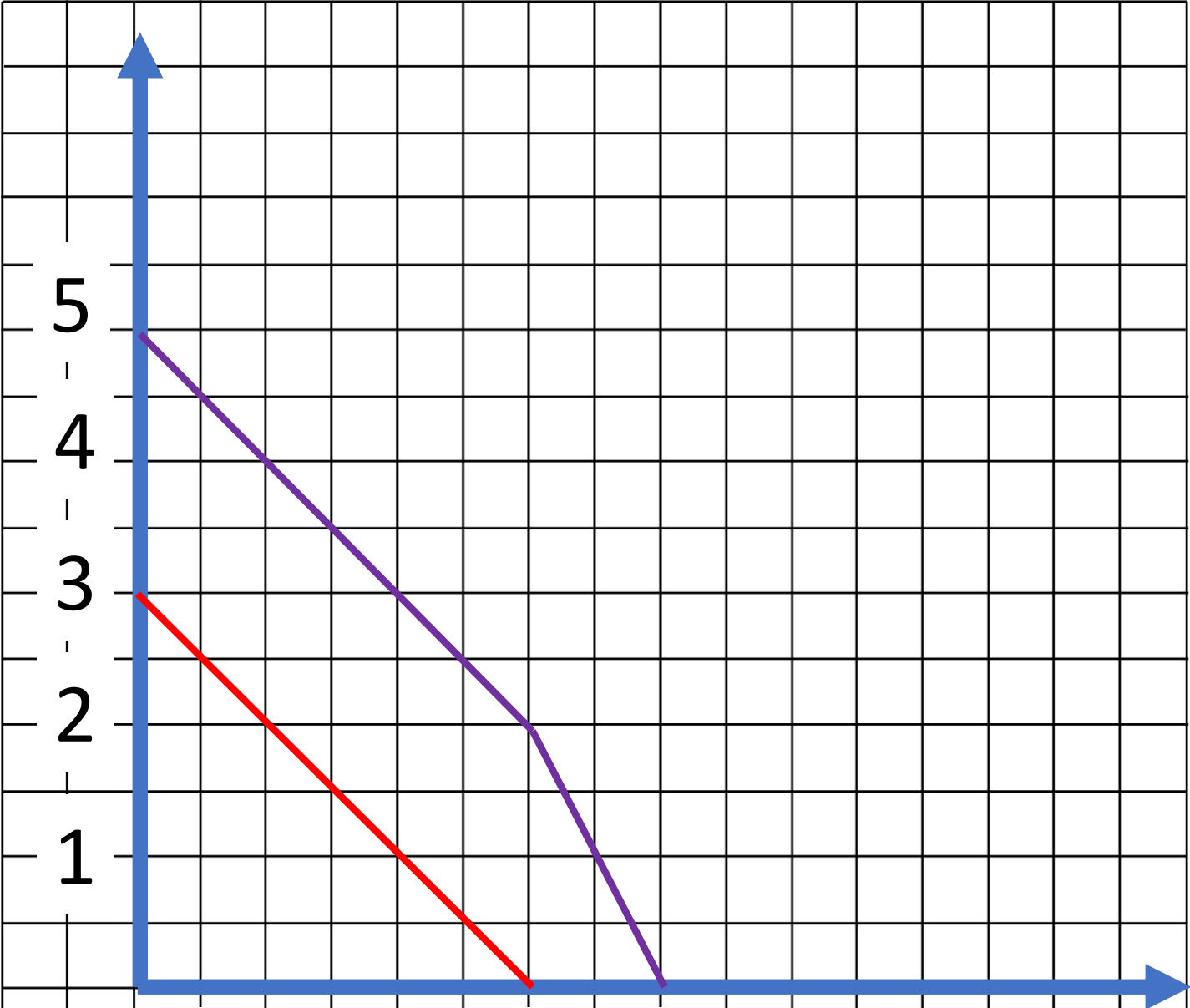
$$= \text{Time to serve the customers in the queue} \\ + \\ \text{Residual service time}$$

Waiting time of an arriving customer

$$= \text{Number of the customers in the queue} * \\ \text{Average service time of each customer} \\ + \\ \text{Residual service time}$$

Time t	Remaining work at time t	
	Job 1	Job 2
0	5	3
1	4.5	2.5
2	4.0	2.0
6	2	0
8	0	0

Remaining work



Time t	Remaining work at time t			
	Job 1	Job 2	Job 3	Job 4
0	5	3		
1	4.5	2.5	4	
2	$4 \frac{1}{6}$	$2.5 - 1/3 = 2 \frac{1}{6}$	$4 - 1/3 = 3 \frac{2}{3}$	1
6	$3 \frac{1}{6}$	$1 \frac{1}{6}$	$2 \frac{2}{3}$	0
$6 + 3 (1 \frac{1}{6})$	2	0	$1 \frac{1}{2}$	

Remaining work

